Implementation Plan: EkanaVet Enterprise Platform

1. Executive Summary

This project aims to build a scalable web platform for a wholesale veterinary company. The system will serve three distinct user groups:

1. **Customers (Clinics/Retailers):** Ordering, payments, appointments, and sample requests.
2. **Employees (Delivery/MRs):** Task management, receipt filing, and commission tracking.
3. **General Public/Users:** AI-driven veterinary advice and location services.

Primary Design Aesthetic: Clean, medical, and trustworthy using a **Green (#2E7D32)** and **White (#FFFFFF)** color palette.

2. Technology Stack & Architecture

To ensure scalability from 6 products to hundreds, we will use a modular architecture.

Frontend (Client-Side)

- **Framework: React.js** (or Next.js for better SEO on the public pages).
- **Styling: Tailwind CSS**. It allows for rapid styling and easy implementation of the Green/White theme with a "utility-first" approach.
- **State Management:** Redux Toolkit or React Query (for handling complex order states and task lists).

Backend (Server-Side)

- **Core API: Node.js with Express** OR **Python (FastAPI/Django)**.
  - *Recommendation:* Python is preferred here because of the heavy AI integration requirements later.
- **Authentication:** Firebase Auth or Auth0 (Handles secure logins for multiple roles: Customer vs. Employee).

Database (Scalable Data)

- **Primary DB: PostgreSQL**. A relational database is essential for connecting Orders -> Customers -> Invoices -> Employee Commissions.
- **File Storage:** AWS S3 or Google Cloud Storage (for storing receipt images uploaded by employees).

AI & Geolocation Services

- **LLM Integration:** OpenAI API (GPT-4) or Google Gemini API.
- **Vector Database:** Pinecone (to store product documentation so the AI knows specifically about *your* 6 products).

- **Maps:** Google Maps Platform (Places API) to find nearby vets/shops.

3. Database Schema Concept (High Level)

This structure ensures that adding new products later is as simple as adding a row to a table.

- **Users Table:** (ID, Name, Email, Role

$$Customer, MR, Delivery, Admin$$

, PasswordHash)
- **Products Table:** (ID, Name, Category, WholesalePrice, StockLevel, Description, *Tag_Embeddings*)
- **Orders Table:** (ID, CustomerID, Status, TotalAmount, PaymentReference)
- **Appointments Table:** (ID, CustomerID, RepID, Date, Type

$$Visit/Virtual$$

)
- **Tasks Table:** (ID, EmployeeID, Type

$$Delivery/Meeting$$

, Status

$$Pending/Done$$

, LocationData)
- **Commissions Table:** (ID, EmployeeID, OrderID, Amount, DateEarned)

4. Feature Breakdown & Logic

A. Customer Portal

1. **E-commerce Engine:**
   - Grid view of products (Scalable: Pagination included).
   - **Payment Gateway:** Integration with Stripe or Razorpay for wholesale bulk payments.
2. **Sample Request System:**
   - A "Request Sample" button distinct from "Add to Cart." Limits quantities per customer to prevent abuse.
3. **Meeting Scheduler:**
   - A calendar interface allowing customers to book time slots with Medical Representatives (MRs).

B. Employee Portal (The "Internal Tool")

1. **Dashboard:**
    - **Delivery Agents:** See a map/list of deliveries for the day. Button to "Mark Delivered" (Updates Order Status in DB).
    - **Medical Reps (MR):** See scheduled visits and sample drop-off tasks.
2. **Financials:**
    - **Upload Receipt:** Mobile-friendly form to snap a photo of fuel/travel receipts.
    - **Commission View:** Real-time calculation of earnings based on completed orders/visits.

C. AI Chatbot (The "Smart Vet Assistant")

This requires a specific workflow called **RAG (Retrieval-Augmented Generation)**.

1. **User Input:** "My cow has a low appetite and fever."
2. **Retrieval:** System searches your product database for medicines matching "appetite" and "fever."
3. **Generation:** AI answers: "Based on symptoms, *Product X* might help. Please consult a vet."
4. **Location:** AI asks for user's zip code -> Hits Google Maps API -> Returns list of nearby clinics.

5. UI/UX Design Guidelines

- **Primary Color:** Emerald Green (`#10B981`) or Forest Green (`#2E7D32`) - Used for Call-to-Action buttons (Buy, Login) and headers.
- **Secondary Color:** White (`#FFFFFF`) - Used for backgrounds to maintain a sterile, clean medical look.
- **Accent Color:** Light Grey (`#F3F4F6`) - For section separation.
- **Typography:** Sans-serif (Inter or Roboto) for readability.

6. Development Phasing (Roadmap)

Phase 1: MVP (Minimum Viable Product)

- **Goal:** Get the products online and allow ordering.
- **Deliverables:**
    - Landing page with company info.
    - Product Catalog (6 items).
    - Shopping Cart & Payment integration.
    - Basic Admin panel to add/edit products.

Phase 2: User Accounts & CRM

- **Goal:** Retain customers and manage interactions.

- **Deliverables:**
  - Customer Login/Registration.
  - Order History.
  - "Request Sample" workflow.
  - Appointment booking system.

Phase 3: Employee Management System

- **Goal:** Manage operations and logistics.
- **Deliverables:**
  - Employee Login (distinct from customers).
  - Task assignment dashboard (Admin assigns tasks -> Employee sees them).
  - Receipt upload & Commission visualization.

Phase 4: AI & Intelligence

- **Goal:** Add value and innovation.
- **Deliverables:**
  - Train the Chatbot on specific animal symptoms and YOUR product interactions.
  - Integrate Google Maps API for the "Find Nearby" feature.

7. Scalability Strategy

- **Product Growth:** The database uses a standard `Product` entity. Adding product #7 or product #700 requires zero code changes, only database entry.
- **Traffic Load:** By using a cloud provider (AWS/Vercel), we can set up auto-scaling groups to handle high traffic during marketing pushes.